US ECONOMICS ANALYST

Productivity Stagnation in the Construction Industry: An International Perspective

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- Between 1970 and 2024, aggregate labor productivity in the US more than doubled. In stark contrast, labor productivity in the construction industry declined by 30% over the same period. The persistent decline in construction productivity appears to be a common trend across advanced economies but it has been most acute in the US, contributing to its housing shortage and affordability crisis. We explore why the US construction industry's productivity has underperformed relative to both the rest of the US economy and other countries.
- One partial explanation is that innovation related to construction has been limited. Most of the major machines used for construction today had already been invented in the 1950s. Since then, technological changes within construction have been incremental, and the industry has also benefitted relatively little from rapid advances in information and communication. Using industry-level patent data, we estimate that technological changes have boosted annual construction productivity growth by 0.8pp since 1965, below the 1-1.3pp boost that we estimate for other industries.
- Increased regulation of housing has also been a major drag on construction productivity growth. New data reveal that land use regulations have gradually become more stringent across major advanced economies over the past 50 years, with regulations tightening the most in the US. Our cross-country analysis shows that regulatory changes lowered annual construction productivity growth by 0.7pp, offsetting much of the boost from improvements in technology and labor quality. For more recent decades, in which more data are available for other advanced economies, changes in land use regulations explain most of the difference in construction productivity growth between US and other G10 countries.
- Looking beyond these fundamental drivers, measurement issues have also likely played a role. In particular, the US construction output price index has likely failed to fully account for improvements in housing quality, resulting in a downward bias to real construction output and productivity. A recent academic study found that properly accounting for quality improvement and several other measurement problems would raise measured annual US construction productivity growth between 1990 and 2024 from the currently reported -1.0% to -0.3%, still well below the 1.7% economy-wide productivity growth rate over

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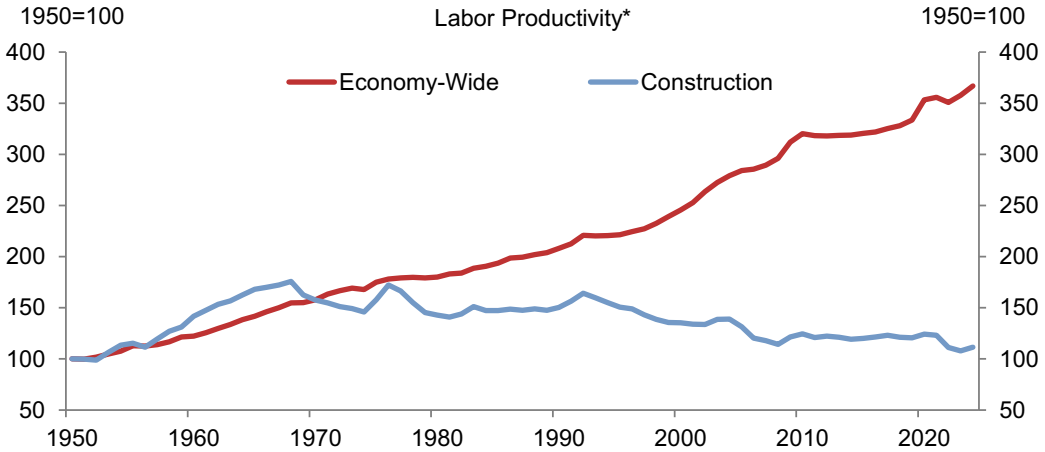
that period. These measurement issues are likely less pronounced for other G10 countries that account for quality improvements using a more thorough bottom-up approach.

- Taken together, our analysis shows that, within the US, the tightening of land use regulations has accounted for 40% of the gap in productivity growth between construction and the rest of the economy since 1965, and the lack of innovation and quality mismeasurement have each accounted for 20%. Looking across countries, we find that more severe tightening of land use regulations and greater quality mismeasurement account for most of the underperformance of the US construction industry since 1991, relative to other major G10 countries.

Productivity Stagnation in the Construction Industry: An International Perspective

Labor productivity in the US construction industry grew steadily from 1950-1970, narrowly outpacing economy-wide productivity growth. But from 1970 to 2024, productivity in the construction industry fell by 30% while economy-wide productivity more than doubled.[1]

Exhibit 1: While Economy-wide Productivity More Than Doubled Between 1970 and 2024, Productivity in the Construction Industry Fell by 30% Over That Period



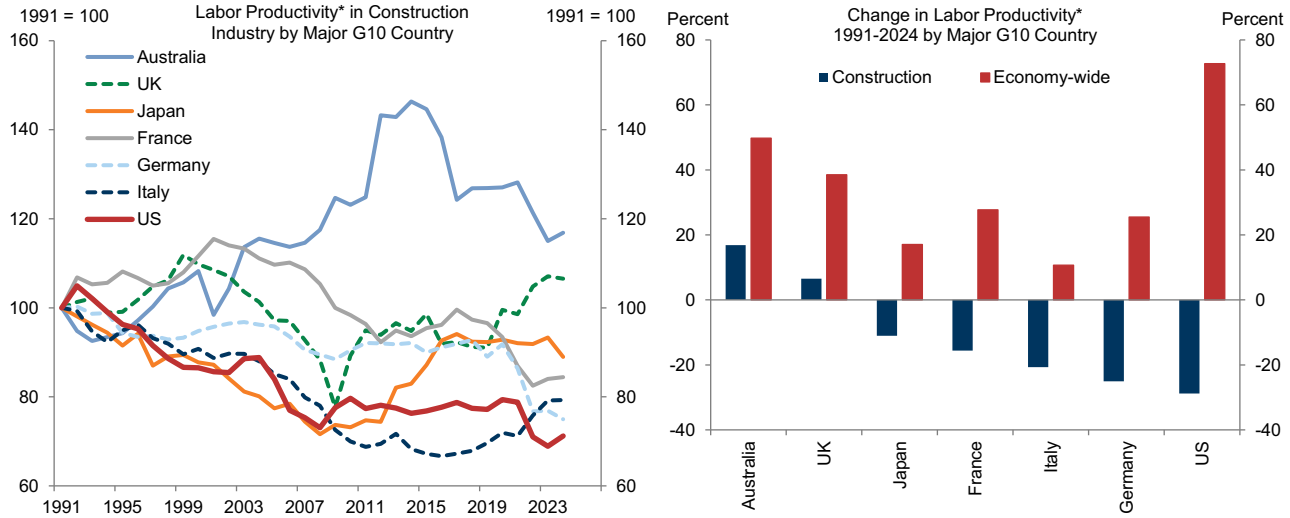
*Labor productivity for 1964-2024 is measured by gross value added per hour worked. For the period 1950-1963, where aggregate hours worked data is unavailable, labor productivity was computed by assuming that hours worked per worker remained at the 1964-1969 average.

Source: Goldman Sachs Global Investment Research, Department of Commerce, Department of Labor

The underperformance of the construction industry is not unique to the US. In the last two decades, during which productivity data are more consistently available for other countries, we find that many other advanced economies also experienced stagnation or persistent decline in construction productivity, despite their economy-wide productivity growing at a solid pace. Among all major G10 countries, however, US construction productivity has experienced by far the largest decline and has also underperformed economy-wide productivity the most.

[1] This trend is also studied by economists Austan Goolsbee and Chad Syverson (2023) and Leonardo D'Amico, Edward Glaeser, Joseph Gyourko, William Kerr, and Giacomo Ponzetto (2024).

Exhibit 2: The Relative Underperformance of Construction Industry Labor Productivity Is a Common Trend Across Advanced Economies, Although It Has Been Most Acute in the US



*Gross value added per hour worked.

Source: Goldman Sachs Global Investment Research, Eurostat, Department of Commerce, Department of Labor

In this week's *US Analyst*, we explore why the US construction industry's productivity has underperformed relative to both the rest of the US economy and other countries.

Driver #1: Lack of Innovation

One partial explanation is that innovation related to construction has been limited. Exhibit 3 shows that most of the major machines used for construction today had already been invented in the 1950s. Since then, technological changes within construction have been incremental.

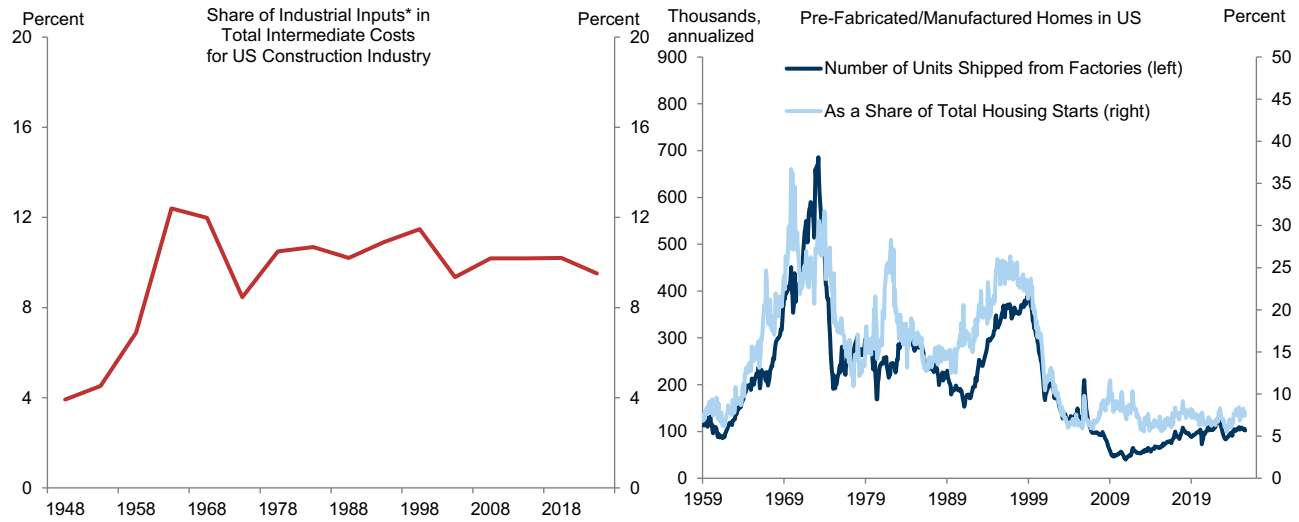
Exhibit 3: Major Machines Used in the Construction Industry Today Had Already Been Invented in the 1940s and 1950s

Type	Usage	Key Innovation Years
Machine & Tool		
Bulldozers	Pushing large quantities of loose materials like soil, sand, and rubble.	1920s
Concrete Mixers	Combining cement, sand, water, and other aggregates.	1930s
Cranes	Lifting and moving heaving materials to different levels of a site.	1940s
Backhoe Loaders	Excavation, trenching, loading materials, and landscaping.	1940s
Excavators	Digging, earthmoving, and demolition.	1950s
Road Rollers	Compacting soil, gravel, or asphalt for building roads, foundations, and other surfaces.	1950s
Nail gun	Driving nails into wood, masonry, or other materials.	1950s
Skid-steer Loaders	Digging, lifting, grading, and snow removal that requires more precision.	1950s
Other		
Off-site fabrication and assembly	Manufacturing and assembling components of buildings in a factory before transporting them to the final site.	1940s, 1970s
Computer-aided design (CAD)	Using software to create 2D drawings and 3D models for building projects.	1960s, 1980s
Industrial robots	Performing on-site tasks with high precision, reducing labor cost, improving safety.	1970s
Information and communication technology	Sourcing inputs and managing projects.	1980s
Drones	Mapping and inspecting sites, monitoring project progress, improving safety.	2000s
3D printing	Building parts of structures with high precision, reducing waste and construction time.	2000s
AI	Analyzing data to provide insights for project management and predict maintenance.	2020s

Source: Goldman Sachs Global Investment Research

The rate of adoption of new technology within construction increased sharply in the 1950s and 1960s but has since stagnated. The left panel of Exhibit 4 shows that the share of industrial machines in total construction production costs increased sharply from 4% in 1948 to 12% in 1968, but the share decreased to only 10% in the 1970s and has remained at that level ever since. The right panel of Exhibit 4 shows that pre-fabrication technology—which helps to accelerate construction by producing components of residential units in off-site facilities—was widely used during 1960-1970. At its peak, it contributed about one-third of the total new residential housing units annually. However, its use has persistently declined since then, now accounting for only 5% of total new residential housing units.

Exhibit 4: The Adoption of Industrial Machines and Pre-Fabrication Technology Increased Sharply Throughout the 1950s and 1960s but Has Since Stagnated

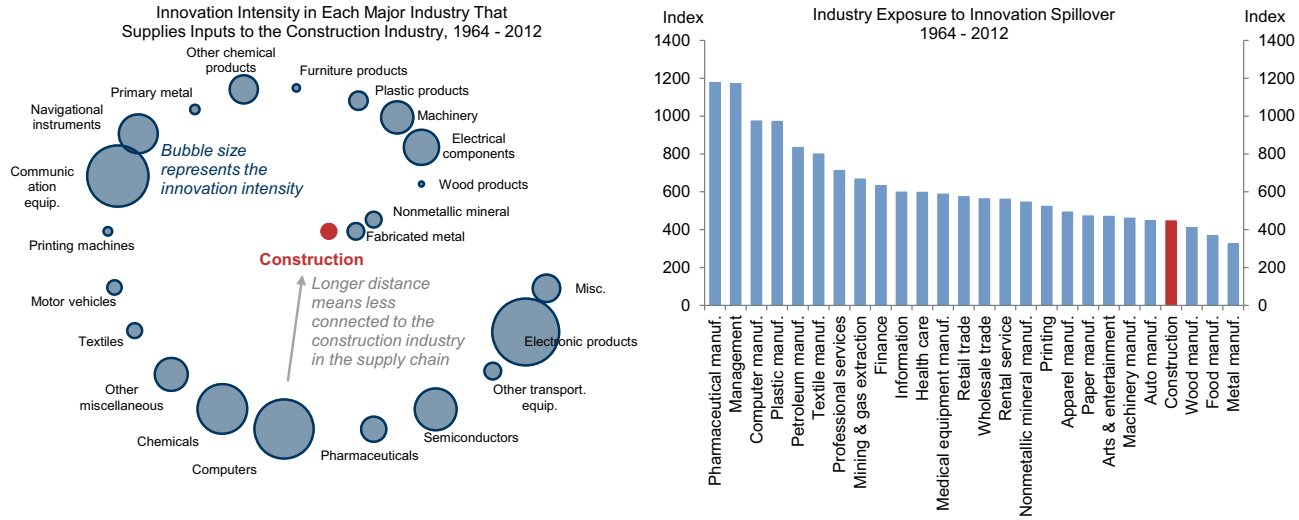


*Industrial machines, electrical equipment, computers, electronics, trucks, and motor vehicles.

Source: Goldman Sachs Global Investment Research, Department of Commerce

In addition to a lack of direct innovation, the construction industry also benefited very little from technological advances in other industries. The left panel of Exhibit 5 shows the intensity of innovation in each major industry, measured by patents per thousand workers from 1964 to 2012, and how closely each of these major industries is linked to the construction industry in the supply chain. The chart shows that major technological breakthroughs over the past 50 years have been concentrated in industries like information, communication, and chemical manufacturing that are quite distant from the construction industry's supply chain. As a result, the economy-wide innovation spillover to the construction industry has remained very limited compared to other industries like pharmaceutical manufacturing and professional management (Exhibit 5, right).

Exhibit 5: The Construction Industry Benefited Very Little from the Innovation Boom in Recent Decades Because Technological Breakthroughs Have Been Concentrated in Industries That Are Distant from the Construction Industry's Supply Chain

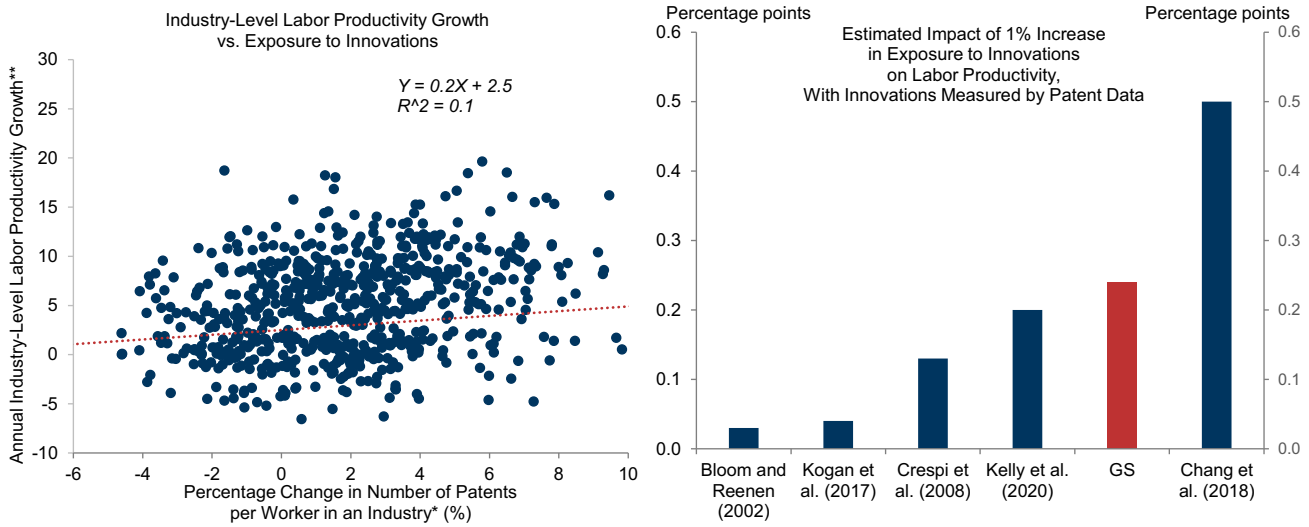


Note: The left panel shows the innovation intensity in each major industry that supplies essential inputs to the construction sector. The size of each blue bubble represents the innovation intensity in an industry, which is defined as the total number of patents filed per thousand workers in that industry from 1964 to 2012. The distance between the central red dot (which symbolizes the construction sector) and any given blue bubble represents the strength of the supply chain connection between the construction industry and that specific input-providing industry. A shorter distance signifies a closer supply chain relationship.
The right panel computes an innovation spillover exposure index for each industry. This index quantifies how much an industry is exposed to, or benefits from, innovation originating from its upstream suppliers. For example, the construction industry's exposure index is computed as a weighted average of the innovation intensities of all other industries that supply inputs to it. The weighting factor is based on the strength of the supply chain connection.

Source: Goldman Sachs Global Investment Research, Department of Commerce, US Patent and Trademark Office

Combining patent data with industry-level productivity growth, we estimate that a 1% increase in industry-level innovation intensity can boost annual productivity growth by about 0.2pp (Exhibit 6, left). This estimate is broadly in line with findings from academic studies that also used patent data to assess the impact of innovation on labor productivity (Exhibit 6, right). Our estimate implies that technological changes have only boosted annual construction productivity by 0.8pp since 1965, below the 1-1.3pp boost we estimate for other industries.

Exhibit 6: An Industry-Level Panel Analysis of Patent and Productivity Data Suggests That a 1% Increase in Innovation Intensity Can Boost Productivity Growth by About 0.2pp



*To measure an industry's exposure to innovation, we consider patents generated directly within this industry and indirectly within industries that provide inputs to this industry.
**The y-axis plots industry's labor productivity in a given year, after netting out industry fixed effects, year fixed effects, and impacts from several other factors that are commonly considered as drivers of productivity (e.g. investment intensity and labor quality).

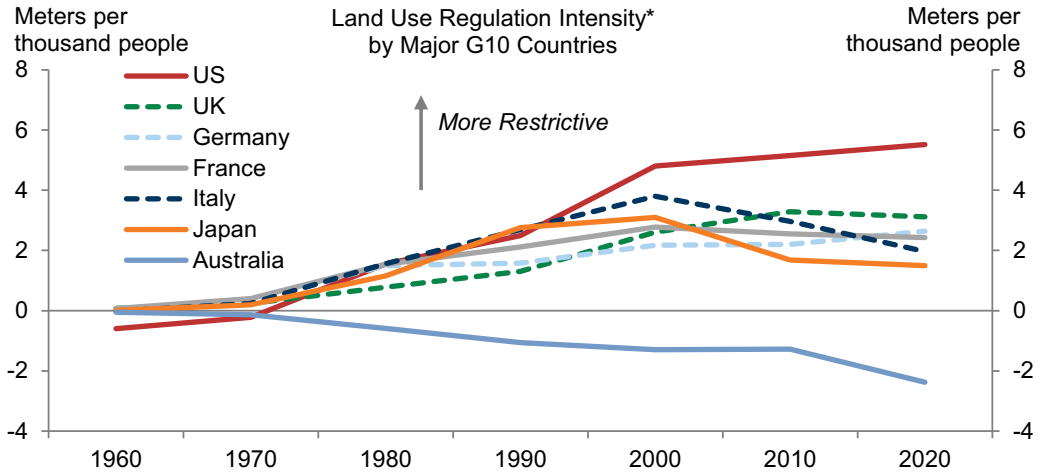
Source: Goldman Sachs Global Investment Research, Department of Commerce, US Patent and Trademark Office

Driver #2: Increased Regulatory Constraints

Increased regulation of housing has also been a major drag on construction productivity growth.

In a prior report, we highlighted that land use regulations in the US have become more stringent over time, and that these restrictive regulations are one of the main drivers of the housing shortage in the US. A new international data set constructed by the economists Remi Jebwab, Jason Barr, and Jan Brueckner (2021) reveals that while land use regulations in major advanced economies were generally very accommodative to local population and demand growth in the 1960s, most of these countries tightened their regulations considerably over the subsequent 50 years. Among all major G10 countries, the US implemented the most stringent regulatory changes.

Exhibit 7: Land Use Regulations Have Gradually Become More Stringent Over Time Across Most Advanced Economies, with Regulations Tightening the Most in the US



*Gap between regulated building height and model-implied latent demand for building height (Jedwab, Barr, and Brueckner 2021).

Source: Goldman Sachs Global Investment Research, Jebwab, Barr, and Brueckner (2021)

Our cross-country analysis further confirms that stricter land use regulations have weighed on construction productivity growth meaningfully (Exhibit 8). On average, we find that a 1% increase in country-level regulation intensity lowers annual construction productivity growth by 0.6pp. But after controlling for other contributors to productivity growth—such as investment intensity, labor quality, and other country-specific characteristics—we find an even larger effect: a 1% increase in country-level regulation intensity lowers construction productivity growth by 0.9pp, holding all else constant.

Our estimate implies that increased regulation in the US since the 1960s has likely reduced annual construction productivity growth by 0.7pp, offsetting much of the estimated 0.8pp boost from technological changes.

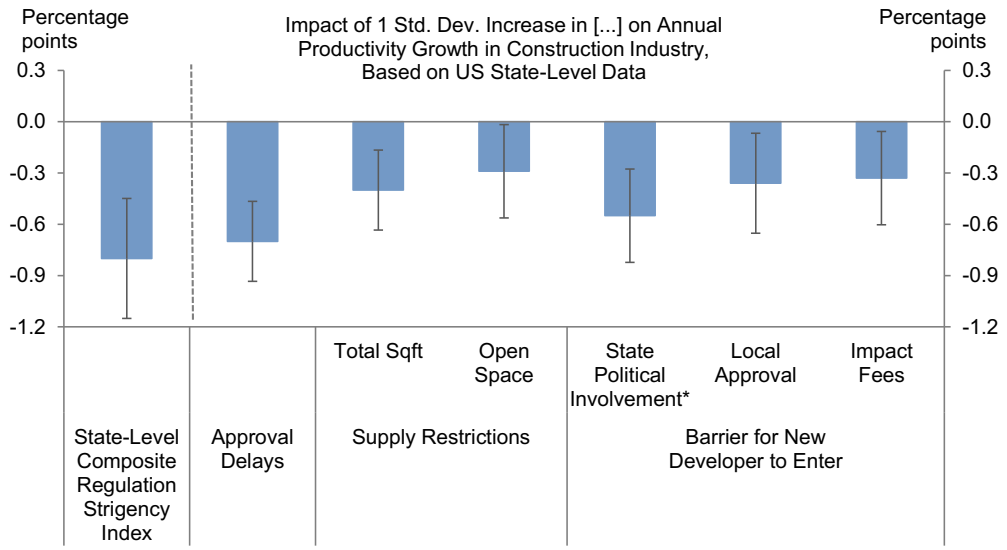
Exhibit 8: Cross-Country Analysis Confirms That Stricter Land-Use Regulation Weighed on Construction Productivity Growth Meaningfully, Even After Controlling for Effects from Other Channels

	Dependent Variable: Annual Construction Productivity Growth (%) 1995-2021		
	(1)	(2)	(3)
Land-Use Regulation Intensity			
1% Increase in Country-level Regulation Intensity Index	**-0.56*** **(0.28)**	**-0.84***** **(0.27)**	**-0.89**** **(0.34)**
Productivity Spillover			
1pp Increase in Productivity Growth of Other Industries Closely Linked to Construction		0.24** (0.09)	0.30** (0.15)
Investment Intensity			
1pp Increase in Investment as a Share of Capital Stock		0.08* (0.04)	0.04 (0.05)
Labor Quality			
1pp Increase in Share of Prime Age Workers in Construction		0.37* (0.22)	0.25 (0.29)
1pp Increase in Share of Workers in Construction With at Least Secondary Degrees		0.02 (0.04)	0.05 (0.05)
Country and Year Fixed Effects	N	N	Y
R-squared	0.1	0.2	0.7

Source: Goldman Sachs Global Investment Research

As noted in our prior work, land use regulations can take many different forms. Using US state-level data, our analysis further shows that approval delays tend to impose the largest drag on construction productivity growth (Exhibit 9). Rules that restrict the size and height of new buildings also weigh on construction productivity growth, likely because they lead to inefficient investment decisions across different locations. State political involvement, local approval requirements, and impact fees can also lower construction productivity growth, by raising barriers for new developers to enter, thereby reducing local competition.

Exhibit 9: Many Types of Regulations Weigh on Construction Productivity Growth, Especially Approval Delays



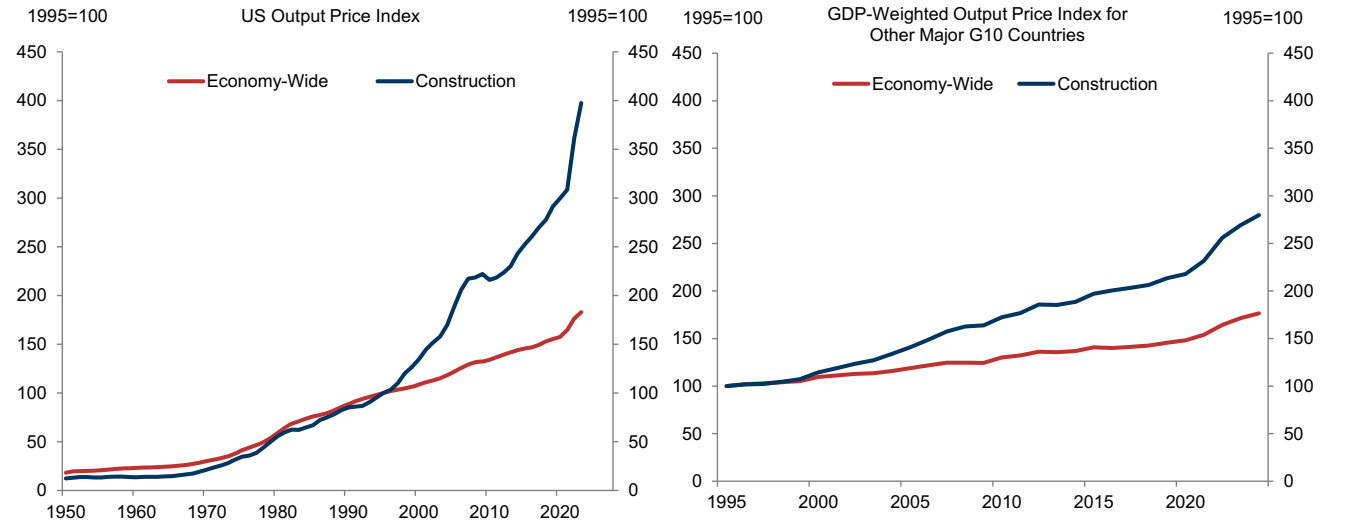
Note: Error bars represent 95% confidence interval.
*State political involvement index measures how involved the state legislature and executive branches are in affecting residential building activities and growth management procedures.

Source: Goldman Sachs Global Investment Research, Department of Commerce, Zell/Lurie Real Estate Center

Driver #3: Mismeasurement in Real Construction Output

Looking beyond these fundamental drivers, measurement issues have also likely played a role. Exhibit 10 shows that the construction output price index has risen at a much faster pace than the aggregate output price index since the 1990s across advanced economies, with the US exhibiting the largest divergence. This raises the concern that the construction output price index might have failed to fully account for improvements in housing quality, resulting in a persistent downward bias to real construction output and productivity.

Exhibit 10: The Construction Output Price Index—Which Has Risen at a Much Faster Pace Than the Aggregate Output Price Index in the Last Two Decades—Has Likely Failed to Account for Improvements in Housing Quality, Resulting in a Persistent Downward Bias to Real Construction Output



Source: Goldman Sachs Global Investment Research, Eurostat, Department of Commerce

In theory, a price index is designed to capture price changes for a specific product over time, while holding the product's features and quality constant. However, constructing a price index for new residential and non-residential structures is especially challenging because they are highly heterogenous and have many characteristics that determine their overall quality. For example, even with identical features, a slight change in the location of the structure would affect the price. There are two commonly used approaches to construct an output price index for new structures. The first is a top-down output price approach, and the second is a bottom-up input price approach.

The US construction output price index relies heavily on the first approach.[2] For new residential units, the statistical agency uses data on final sales prices of new residential housing units and a set of observed characteristics (summarized in the first column of Exhibit 11). These characteristics are used to ensure that housing units of the "same quality" are consistently tracked over time. However, over the past few decades, there have been meaningful quality improvements in areas such as construction materials, electrical systems, and heating and air conditioning, which are not adequately captured by the current estimation strategy (the second column of Exhibit 11).

[2] See Census' methodology documentation for the new single-family house price index for more information.

Exhibit 11: The US Construction Price Index for New Residential Housing Units Likely Missed Some Quality Improvements, Leading to a Downward Bias to Real Construction Output and Productivity

Characteristics Likely Affecting New Residential Home Price	
Characteristics Used to Construct the Index	**Quality Improvements Likely Not Captured**
• Size of house (floor area) • Geographic location • Number of bedrooms • Number of bathrooms • Number of fireplaces • Number of parking gararages • Basement (yes/no) • Deck (yes/no) • Patio (yes/no) • Construction method (4 types) • Type of exterior wall material (11 types) • Type of heating/air conditioning (7 types)	• Heating/air conditioning becomes more energy efficient • Better insulation • Improvement in electrical system • Higher-quality materials • Reinforced concrete for foundation • Improvement in water and waste management

Source: Goldman Sachs Global Investment Research

A recent academic study by Fed economists Daniel Garcia and Raven Molloy (2023) found that failure to account for these quality changes likely resulted in a 0.5pp drag on annual construction productivity growth in the US since 1990. In addition, the official statistics also use the residential housing price index as a deflator for nonresidential structures (Exhibit 12), which likely contributed to an additional 0.2pp drag on annual construction productivity growth. This means that in the absence of these mismeasurement issues, average annual labor productivity growth in the US construction industry from 1990 to 2024 would have been -0.3% instead of -1.0%.

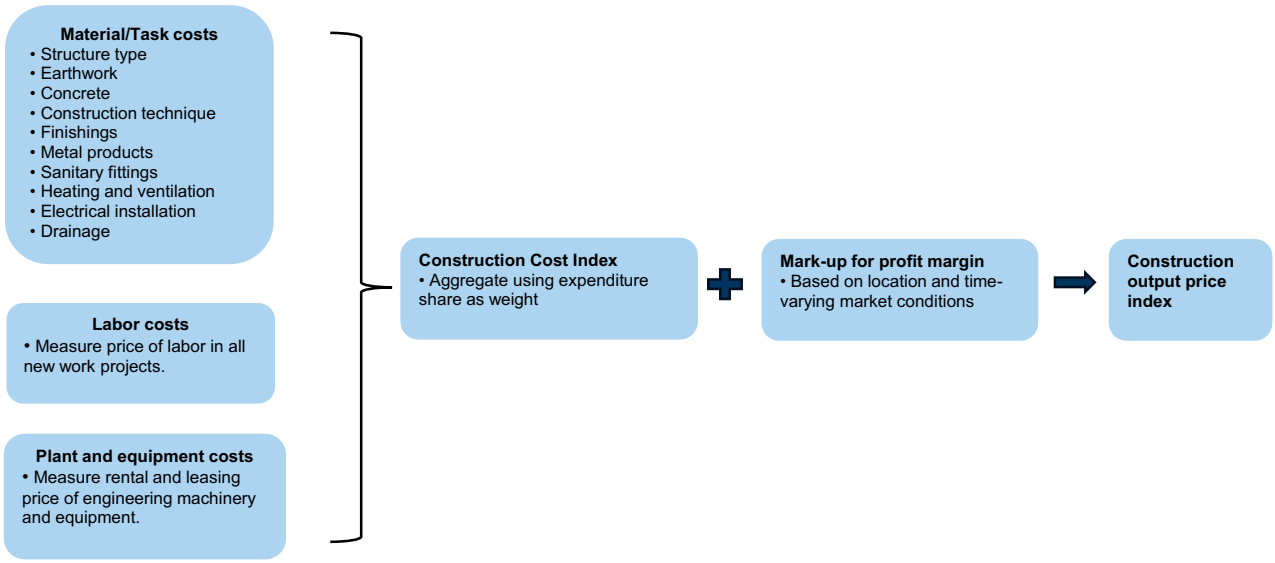
Exhibit 12: The Residential Housing Price Index Is Inappropriately Used as a Deflator for Nonresidential Structures, Leading to an Additional Downward Bias to Real Construction Output and Productivity

Price Deflator for Construction Output, by Type		
Construction output type	**Price index**	**Nominal output share (%)**
New single-family residential units	Price index for new single-family homes	31.4
New multifamily residential units	Price index for new multifamily homes	4.9
Residential improvements	Weighted average of new single-family price indices, PPI for inputs to residential maintenance and repair, and employment cost index for the construction industry	19.8
Nonresidential sectors	Before 1997: Simple average of price index for single-family homes and building cost index produced by the Turner Construction Company After 1997: Sector-specific PPI or cost index based on various construction costs	43.9

Source: Goldman Sachs Global Investment Research

Unlike the US, Japan and many countries in Europe rely more on the bottom-up input approach. This approach aggregates prices of inputs used for construction to form a construction cost index. It then overlays a profit margin on top of this cost index to form the final construction output price index. While no comprehensive studies attempt to quantify the mismeasurement bias for these countries, by considering the specific types of material and engineering machinery inputs used, this bottom-up approach likely better captures the nuances of quality improvements over time than the top-down approach used by the US.

Exhibit 13: Japan and Many Countries in Europe Rely More on a Bottom-Up Input Approach to Estimate a Construction Output Price Index, Which Likely Better Captures Quality Improvements Over time



Source: Goldman Sachs Global Investment Research

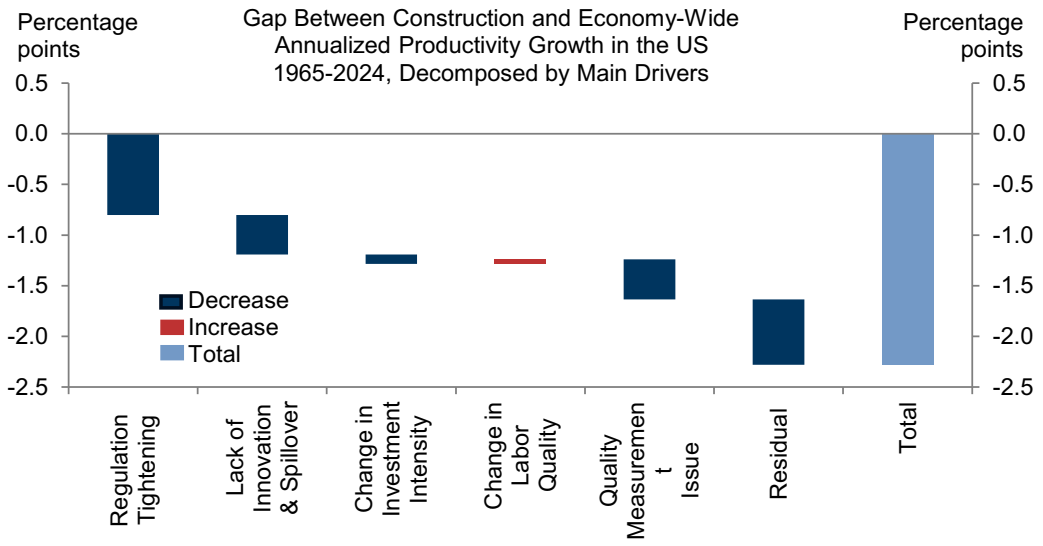
Although not all the differences in the cumulative inflation gap between construction and the rest of the economy across countries can be attributed to methodological differences (Exhibit 10), the distinct approaches used by the US and other countries, as outlined in Exhibits 11-13, lead us to suspect that mismeasurement bias accounts for some of this gap.

Conclusion

Since 1965, US construction labor productivity has been falling at an average pace of 0.6% per year, while economy-wide productivity has been growing at about 1.6% per year.[3] Our analysis shows that the tightening of land use regulations has accounted for 40% of the gap in productivity growth between construction and the rest of the economy, and the lack of innovation and quality mismeasurement have each accounted for 20%.

[3] For the US, we use gross value-added per hour worked to measure economy-wide labor productivity so that it is consistent with labor productivity reported by other OECD countries. The more commonly reported labor productivity measure for the US—real nonfarm business value-added output per hour worked—has been growing at about 2% per year since 1965.

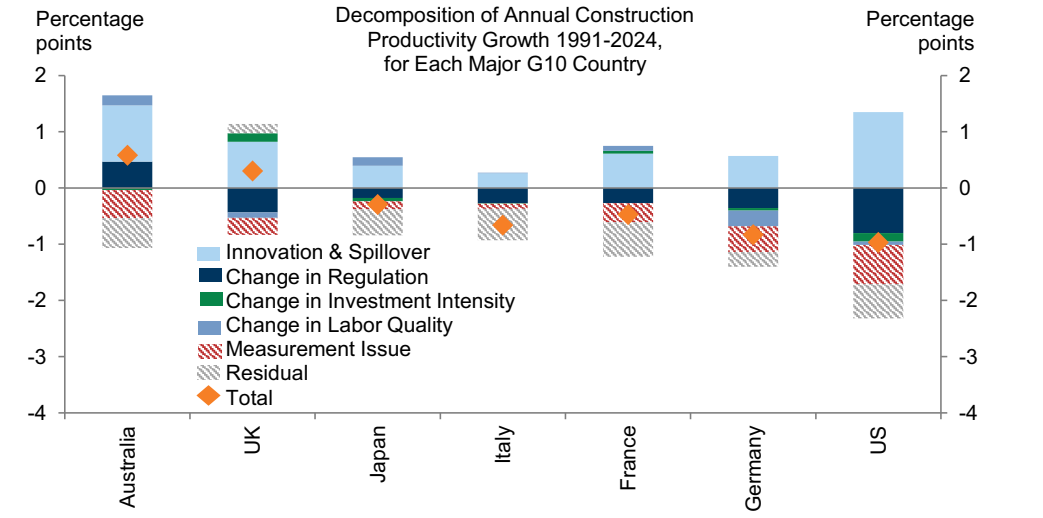
Exhibit 14: Tightening of Land Use Regulations Has Accounted for 40% of the Gap Between Construction and Economy-Wide Productivity Growth in the US Since 1965, and Lack of Innovation and Quality Mismeasurement Issue Have Each Accounted for 20%



Source: Goldman Sachs Global Investment Research

Looking across major G10 countries for the period of 1991-2024, US construction productivity growth averaged about -1.0% per year, vs. -0.4% for G10 European countries and Japan, and +0.6% for Australia. More severe tightening of land use regulations and greater quality mismeasurement account for most of the underperformance of the US.

Exhibit 15: Tightening of Land Use Regulations and Quality Mismeasurement Issues Helped to Explain Most of the Underperformance in US Construction Productivity Growth, Relative to Other G10 Countries



Note: Shaded bars indicate estimates with lower confidence.

Source: Goldman Sachs Global Investment Research

Elsie Peng

The US Economic and Financial Outlook

THE US ECONOMIC AND FINANCIAL OUTLOOK
(% change on previous period, annualized, except where noted)

	2023	2024	2025	2026	2027	2025 Q1	2025 Q2	2025 Q3	2025 Q4	2026 Q1	2026 Q2	2026 Q3	2026 Q4
OUTPUT AND SPENDING													
Real GDP	**2.9**	**2.8**	2.2	2.8	2.1	**-0.6**	**3.8**	**4.4**	2.0	3.1	2.6	2.1	2.1
Real GDP (annual=Q4/Q4, quarterly=yoy)	**3.4**	**2.4**	2.4	2.5	2.1	**2.0**	**2.1**	**2.3**	2.4	3.4	3.1	2.5	2.5
Consumer Expenditures	**2.6**	**2.9**	2.7	2.4	2.1	**0.6**	**2.5**	**3.5**	2.6	2.1	2.4	2.1	2.1
Residential Fixed Investment	**-7.8**	**3.2**	-2.4	-2.3	2.0	**-1.0**	**-5.1**	**-7.1**	-5.0	-3.0	1.5	2.0	2.0
Business Fixed Investment	**7.3**	**2.9**	4.2	5.0	4.1	**9.5**	**7.3**	**3.2**	3.9	5.8	5.8	5.0	4.4
Structures	**16.7**	**1.1**	-4.6	2.4	3.2	**-3.1**	**-7.5**	**-5.0**	5.3	4.0	5.0	4.0	3.0
Equipment	**2.9**	**3.5**	8.2	5.6	3.8	**21.3**	**8.5**	**5.2**	3.1	6.5	6.5	5.5	4.5
Intellectual Property Products	**6.2**	**3.5**	5.6	5.8	4.7	**6.5**	**15.0**	**5.6**	4.0	6.0	5.5	5.0	5.0
Federal Government	**3.3**	**3.8**	-1.2	1.6	0.9	**-5.6**	**-5.3**	**2.7**	-17.5	22.5	0.0	1.0	1.0
State & Local Government	**3.6**	**3.8**	2.5	0.8	1.2	**1.9**	**3.1**	**2.0**	0.5	0.3	0.3	0.5	1.0
Net Exports ($bn, '17)	**-925**	**-1,033**	-1,073	-953	-997	**-1,381**	**-1,058**	**-956**	-896	-929	-948	-962	-975
Inventory Investment ($bn, '17)	**47**	**44**	32	20	61	**172**	**-18**	**-24**	0	5	25	25	25
Nominal GDP	**6.7**	**5.3**	5.1	5.7	4.3	**2.9**	**6.0**	**8.3**	5.2	6.4	5.1	4.1	3.8
Industrial Production, Mfg.	**-1.0**	**-1.0**	1.0	2.5	3.2	**4.0**	**2.5**	**2.8**	**-0.8**	3.4	3.9	3.4	3.2
HOUSING MARKET													
Housing Starts (units, thous)	**1,421**	**1,371**	1,339	1,286	1,322	**1,401**	**1,354**	**1,339**	1,260	1,280	1,290	1,280	1,292
New Home Sales (units, thous)	**665**	**685**	674	670	644	**655**	**665**	**696**	680	690	690	662	638
Existing Home Sales (units, thous)	**4,103**	**4,067**	4,084	4,256	4,420	**4,127**	**3,990**	**4,020**	**4,200**	4,212	4,232	4,267	4,311
Case-Shiller Home Prices (%yoy)*	**5.3**	**3.8**	0.4	0.8	2.2	**3.9**	**2.4**	**1.5**	0.4	0.1	0.6	0.7	0.8
INFLATION (% ch, yr/yr)													
Consumer Price Index (CPI)**	**3.3**	**2.9**	**2.7**	2.0	2.0	**2.7**	**2.5**	**2.9**	**2.8**	2.5	2.7	2.4	2.1
Core CPI **	**3.9**	**3.2**	**2.6**	2.0	2.0	**3.1**	**2.8**	**3.1**	**2.7**	2.5	2.6	2.2	2.1
Core PCE** †	**3.1**	**3.0**	**3.0**	2.1	2.0	**2.8**	**2.7**	**2.9**	2.8	2.8	2.7	2.5	2.3
LABOR MARKET													
Unemployment Rate (%)^	**3.8**	**4.1**	**4.4**	4.5	4.4	**4.2**	**4.1**	**4.4**	**4.4**	4.5	4.5	4.5	4.5
U6 Underemployment Rate (%)^	**7.2**	**7.6**	**8.4**	8.7	8.6	**7.9**	**7.7**	**8.1**	**8.4**	8.6	8.7	8.7	8.7
Payrolls (thous, monthly rate)	**216**	**168**	**49**	64	90	**111**	**55**	**51**	**-22**	53	62	70	70
Employment-Population Ratio (%)^	**60.1**	**59.9**	**59.7**	59.5	59.3	**59.9**	**59.7**	**59.7**	**59.7**	59.6	59.5	59.5	59.5
Labor Force Participation Rate (%)^	**62.5**	**62.5**	**62.4**	62.3	62.1	**62.5**	**62.3**	**62.5**	**62.4**	62.4	62.3	62.3	62.3
Average Hourly Earnings (%yoy)	**5.5**	**4.6**	**4.0**	3.9	3.6	**4.2**	**4.2**	**3.9**	**3.8**	3.9	3.9	3.8	3.8
GOVERNMENT FINANCE													
Federal Budget (FY, $bn)	**-1,694**	**-1,833**	**-1,775**	2,050	2,200	--	--	--	--	--	--	--	--
FINANCIAL INDICATORS													
FF Target Range (Bottom-Top, %)^	**5.25-5.5**	**4.25-4.5**	**3.5-3.75**	3-3.25	3-3.25	**4.25-4.5**	**4.25-4.5**	**4-4.25**	**3.5-3.75**	3.5-3.75	3.25-3.5	3-3.25	3-3.25
10-Year Treasury Note^	**3.88**	**4.58**	**4.18**	4.20	4.25	**4.23**	**4.24**	**4.16**	**4.18**	4.20	4.20	4.20	4.20
Euro (€/$)^	**1.11**	**1.04**	**1.17**	1.22	1.24	**1.08**	**1.18**	**1.17**	**1.17**	1.17	1.21	1.21	1.22
Yen ($/¥)^	**141**	**157**	**157**	153	139	**150**	**144**	**148**	**157**	157	154	154	153

* Weighted average of metro-level HPIs for 381 metro cities where the weights are dollar values of housing stock reported in the American Community Survey. Annual numbers are Q4/Q4.
** Annual inflation numbers are December year-on-year values. Quarterly values are Q4/Q4.
† PCE = Personal consumption expenditures. ^ Denotes end of period.
Note: Published figures in bold.

Source: Goldman Sachs Global Investment Research

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